Talagrand's discrete-convexity conjecture

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Abstract

We prove Talagrand's discrete-convexity conjecture. There is a universal integer k such that, whenever an arbitrary family has Bernoulli product measure at least $1 - 1/k$, the sets not contained in a union of k members admit a cover of total cost at most $1/2$ at the same density.

1 Introduction

Setting. Let $[N] = \{1, \dots, N\}$, and let μ_p be the probability measure on $2^{[N]}$ under which each element is present independently with probability $p \in (0, 1)$. For $\mathcal{D} \subseteq 2^{[N]}$ and an integer $k \geq 1$, the *exceptional family* is

$$E_k(\mathcal{D}) = \{S \subseteq [N] : \text{no } D_1, \dots, D_k \in \mathcal{D} \text{ satisfy } S \subseteq D_1 \cup \cdots \cup D_k\}. \tag{1.1}$$

The D_j may repeat and need not be disjoint. Two immediate facts: $E_k(\mathcal{D})$ is closed under supersets, and $\varnothing \notin E_k(\mathcal{D})$ when $\mathcal{D} \neq \varnothing$.

A family $\mathcal{G}$ of sets *covers* a family $\mathcal{A}$ if every $S \in \mathcal{A}$ contains some $I \in \mathcal{G}$. For $\rho \in (0, 1)$, the *cost* of $\mathcal{G}$ at density ρ is $c_\rho(\mathcal{G}) = \sum_{I \in \mathcal{G}} \rho^{|I|}$, and $\mathcal{A}$ is *ρ-small* if it has a cover $\mathcal{G}$ with $c_\rho(\mathcal{G}) \leq 1/2$. Since $\mu_\rho\{S : I \subseteq S\} = \rho^{|I|}$, a union bound shows that a ρ-small family has μ_ρ-measure at most $1/2$; the cover is an explicit certificate of this.

Theorem 1.1. *Set $k = 2^{75}$. For every integer $N \geq 1$, every $p \in (0, 1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) \geq 1 - \frac{1}{k} \quad \Longrightarrow \quad E_k(\mathcal{D}) \text{ is } p\text{-small.}$$

The constant is independent of N, p and $\mathcal{D}$, and $\mathcal{D}$ is arbitrary (no monotonicity). This is Talagrand's conjecture [11, Conjecture 7.1], restated as Research Problem 13.3.2 in [12], whose Proposition 13.3.3 explains its connection to positive selector processes. The name reflects the analogy with his Gaussian convexity question [11, Conjecture 2.1], where sums of a balanced set play the role of unions.

Proof at a glance. The logical order is Lemma 2.1 ⇒ Proposition 3.1 ⇒ Theorem 1.1.

- *Weights (Section 3, Steps 1–3).* Let $\mu_q(\mathcal{F}) \geq 1/2$. The biased Fourier expansion of $\mathbf{1}_\mathcal{F}$ yields weights $w(U) \geq 0$, $U \subseteq [N]$, with a global bound $\sum_U q^{|U|} w(U) \leq 1$ (Parseval) and, for every $S \in E_{32}(\mathcal{F})$, a local bound $\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \geq 2^{-32}$. The local bound comes from a signed product measure on $32 \times N$ arrays that vanishes unless the 32 rows jointly contain S.

- *From weights to a cover (Section 3, Steps 4–7).* Sets with $w(U) \geq 2^{-64|U|}$ are used as generators; the global bound makes them cheap at density $q/2^{70}$. The remaining sets are sorted into bins by size r and weight level $w(U) \approx 2^{-h}$. If S contains fewer than 2^{12h} sets from each bin, then each bin contributes in total less than $2^{12h} \cdot 2^{-16h} = 2^{-4h}$ to the local sum, which then falls below 2^{-32}. So an exceptional set that avoids the first generators contains at least 2^{12h} sets of some bin.
- *Covering lemma (Section 2).* Lemma 2.1 covers, cheaply and by a random construction, all sets containing 2^{12h} members of one bin. This yields Proposition 3.1: $E_{32}(\mathcal{F})$ is $(q/2^{70})$-small.
- *Density recovery (Section 4).* With $L = 2^{70}$, couple L random sets, each of law μ_p, whose union has law μ_{Lp}. The family $\mathcal{F}$ of unions of L members of $\mathcal{D}$ then has $\mu_{Lp}(\mathcal{F}) \geq 1/2$ and $E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D})$, so the proposition with $q = Lp$ gives density $q/L = p$.

The rest of this section is not used in the proof of Theorem 1.1.

A consequence: positive selectors. (The source's family T is renamed $\mathcal{T}$; T is a number in Section 2.) Following [11, Section 7], let $\mathcal{T} \subseteq [0,\infty)^N$ be nonempty,

$$\phi(S) = \sup_{\tau \in \mathcal{T}} \sum_{i \in S} \tau_i, \qquad M = \mathbb{E}_{\mu_p} \phi,$$

and suppose $0 < M < \infty$. Then $\{S : \phi(S) \geq k^2 M\}$ is p-small for $k = 2^{75}$; the source notes that this recovers the conclusion of Park and Pham [10, Theorem 1.2]. Proof: put $\mathcal{D} = \{D : \phi(D) < kM\}$. Markov's inequality gives $\mu_p(\mathcal{D}) \geq 1 - 1/k$. Since the τ_i are nonnegative, ϕ is monotone and $\phi(A \cup B) \leq \phi(A) + \phi(B)$ (bound each τ separately, then take suprema). Hence $S \subseteq D_1 \cup \cdots \cup D_k$ with all $D_j \in \mathcal{D}$ forces $\phi(S) < k \cdot kM$, so $\{\phi \geq k^2 M\} \subseteq E_k(\mathcal{D})$, and the cover from Theorem 1.1 covers it. This gives a covering certificate, not just the measure bound that Markov's inequality provides.

Related work. (Attributions as given in the source.) Hua, Song and Tudose resolve the Gaussian problem and derive a discrete statement with three unions and covering density p^C [5, Corollaries 1.3–1.4]. Ascoli, He, Park and Talagrand give a threshold reformulation and prove graph-containment cases [1, Proposition 1.5 and Theorems 1.10–1.13]. Talagrand introduced a fractional notion of smallness [11, Sections 6 and 7], defined below; Li resolves the fractional problem [6, Theorem 1.9], and Park derives an integral cover of $E_2(\mathcal{D})$ at density $p/(C \max\{1, \log\log(1/p)\})$ when $\mu_p(\mathcal{D}) > 1/2$ [9, Corollary 3.5]. On methods: with two rows, the signed measure of Section 3 is, per the source, Li's signed kernel multiplied by the Bernoulli masses [6, Section 2]; Park notes the extension to an even number of unions [9, Corollary 3.5, footnote 3]; signed product operators and biased Fourier analysis go back to [4, Section 2.2] and [2, Sections 2.2 and 4.1]; random unions also appear in the covering constructions of [3, Sections 1.1 and 2].

A two-union corollary. This corollary trades fewer unions for a constant density loss and rests on two cited results. Call $\mathcal{A}$ *fractionally ρ-small* if there are coefficients $a_I \geq 0$ ($I \subseteq [N]$) with

$$\sum_{I \subseteq S} a_I \geq 1 \quad (S \in \mathcal{A}), \qquad \sum_{I \subseteq [N]} \rho^{|I|} a_I \leq \frac{1}{2}.$$

(The source displays this at $\rho = p$ and uses it at other densities.)

Corollary 1.2 (Two unions at constant density loss)**.** *There is a universal constant $C \geq 1$ such that, for every integer $N \geq 1$, every $p \in (0,1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) > \frac{1}{2} \quad \Longrightarrow \quad E_2(\mathcal{D}) \text{ is } (p/C)\text{-small.}$$

Proof. We write σ for the density the source calls r.

1. If $E_2(\mathcal{D}) = \varnothing$, the empty cover suffices. Otherwise $E_2(\mathcal{D})$ is nonempty, increasing, and proper: it omits $\varnothing$, since $\mu_p(\mathcal{D}) > 1/2$ gives $\mathcal{D} \neq \varnothing$.
2. *Cited.* By Li's result [6, Corollary 3.1] in Park's fractional-cover formulation [9, Theorem 3.4], $E_2(\mathcal{D})$ is fractionally σ-small for $\sigma = p/2$.
3. Replace each a_I by $\min\{a_I, 1\}$. The cost does not increase. Coverage persists: if some $I \subseteq S$ had $a_I > 1$, that term alone is now 1; otherwise nothing changed for S.
4. *Cited.* The companion expectation-threshold theorem [8, Theorem 1.1] (called the "constant-factor rounding theorem" in the source's introduction) then gives an integral cover of cost at most $1/2$ at density $\sigma/(25 \cdot 512^4) = p/(50 \cdot 512^4)$. Thus $C = 50 \cdot 512^4$ suffices. □

Unverified step. Steps 2 and 4 quote [6], [9] and [8] exactly as the source uses them. The source states neither their hypotheses nor the origin of $\sigma = p/2$, and I could not check them against the cited papers. Steps 1 and 3 and the arithmetic are verified.

2 An elementary covering lemma

The lemma is applied in Section 3, once per bin, with $X = [N]$. Covers and costs are defined as in Section 1 with X in place of $[N]$. An *edge* is a member of $\mathcal{H}$.

Lemma 2.1. *Let X be a finite set, let $0 < \rho < 1$, and let r, h be integers with $1 \leq r \leq h$ and $h \geq 64$. Suppose that $\mathcal{H} \subseteq \binom{X}{r}$ satisfies*

$$|\mathcal{H}|\rho^r \leq 2^{h+1}.$$

Then the family

$$\left\{S \subseteq X : |\{e \in \mathcal{H} : e \subseteq S\}| \geq 2^{12h}\right\}$$

has a cover $\mathcal{G} \subseteq 2^X$ with

$$c_\rho(\mathcal{G}) \leq 3 \cdot 2^{-h}.$$

Proof. Plan. We build $\mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2$, where $\mathcal{C}_0$ is deterministic and $\mathcal{C}_1, \mathcal{C}_2$ depend on a random sample. We show that $\mathcal{G}$ covers the target family for every outcome (Step 4) and that $\mathbb{E}[c_\rho(\mathcal{G})] \leq 3 \cdot 2^{-h}$ (Steps 1, 3, 6). Throughout,

$$m = 2^{12h}, \qquad T = 2^{4h}, \qquad \pi = 2^{-8h}.$$

Step 1: sets of large weighted degree. For $I \subseteq X$ with $|I| \leq r$ put

$$d(I) = |\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|}, \qquad \mathcal{C}_0 = \{I \subseteq X : |I| \leq r,\ d(I) > T\}.$$

Since $d(I)\rho^{|I|} = |\{e \in \mathcal{H} : I \subseteq e\}|\rho^r$ and each edge has 2^r subsets, $\sum_{|I| \leq r} d(I)\rho^{|I|} = 2^r|\mathcal{H}|\rho^r$. Each $I \in \mathcal{C}_0$ satisfies $\rho^{|I|} < T^{-1}d(I)\rho^{|I|}$, so, by the hypothesis on $|\mathcal{H}|\rho^r$ and $r \leq h$,

$$c_\rho(\mathcal{C}_0) \leq T^{-1}2^r|\mathcal{H}|\rho^r \leq 2^{-4h}\, 2^r\, 2^{h+1} = 2^{r+1-3h} \leq 2^{1-2h}. \tag{2.1}$$

Step 2: regular edges and the extension bound. Call an edge *regular* if it contains no member of $\mathcal{C}_0$, and let $\mathcal{H}'$ be the set of regular edges. Thus $d(I) \le T$ whenever I is a subset of a regular edge. We claim that, for every $B \subseteq X$,

$$\sum_{e \in \mathcal{H}'} \rho^{|e \setminus B|} \le T \sum_{i=0}^{r} \binom{|B|}{i}. \tag{2.2}$$

Group the edges $e \in \mathcal{H}'$ by $I = e \cap B$, a subset of B of size at most r; there are at most $\sum_{i=0}^{r} \binom{|B|}{i}$ groups. Every edge in the group of I has $|e \setminus B| = r - |I|$, so a nonempty group contributes

$$|\{e \in \mathcal{H}' : e \cap B = I\}|\, \rho^{r-|I|} \le d(I) \le T,$$

the last step because I lies in a regular edge.

Step 3: sampled pairs. Independently for each ordered pair $(e, e') \in (\mathcal{H}')^2$, diagonal pairs included, *sample* the pair with probability π. Let $\mathcal{C}_1$ be the set of unions $e \cup e'$ of sampled pairs. By (2.2) with $B = e$ (so $|B| = r$ and the binomial sum is 2^r), $\sum_{e' \in \mathcal{H}'} \rho^{|e' \setminus e|} \le 2^r T$. Using $|e \cup e'| = r + |e' \setminus e|$ and linearity of expectation (a union arising from several pairs is counted several times, which only increases the bound),

$$\mathbb{E}[c_\rho(\mathcal{C}_1)] \le \pi \sum_{e \in \mathcal{H}'} \rho^r \sum_{e' \in \mathcal{H}'} \rho^{|e' \setminus e|} \le \pi |\mathcal{H}'| \rho^r 2^r T \le 2^{-8h}\, 2^{h+1}\, 2^r\, 2^{4h} \le 2^{1-2h}. \tag{2.3}$$

Step 4: residual generators; $\mathcal{G}$ is a cover. For every ordered m-tuple $(e_1, \dots, e_m)$ of distinct regular edges such that none of the pairs (e_i, e_j), $1 \le i, j \le m$, was sampled, put $e_1 \cup \cdots \cup e_m$ into $\mathcal{C}_2$. Now fix any outcome and let S contain at least m edges but no member of $\mathcal{C}_0$. Every edge $e \subseteq S$ is regular, since a member of $\mathcal{C}_0$ inside e would lie inside S. Pick m distinct edges $e_1, \dots, e_m \subseteq S$. If some pair (e_i, e_j) was sampled, then $e_i \cup e_j \in \mathcal{C}_1$ lies in S; otherwise $e_1 \cup \cdots \cup e_m \in \mathcal{C}_2$ lies in S.

Step 5: total weight of tuples. For $0 \le t \le m$ let

$$W_t = \sum_{\substack{(e_1, \dots, e_t) \in (\mathcal{H}')^t \\ \text{distinct}}} \rho^{|e_1 \cup \cdots \cup e_t|}, \qquad W_0 = 1.$$

We claim $W_{t+1} \le T(1 + rm)^r W_t$ for $t < m$. Fix a t-tuple with union B; then $|B| \le rt \le rm$. Its extensions are indexed by regular edges e distinct from $e_1, \dots, e_t$; summing over all regular e only increases the sum. Since $\rho^{|B \cup e|} = \rho^{|B|} \rho^{|e \setminus B|}$, (2.2) gives

$$\sum_{e \in \mathcal{H}'} \rho^{|B \cup e|} \le \rho^{|B|}\, T \sum_{i=0}^{r} \binom{|B|}{i} \le \rho^{|B|}\, T (1 + rm)^r.$$

The last step is $\sum_{i=0}^{r} \binom{n}{i} \le (1+n)^r$: list a subset of size at most r of an ordered n-set in increasing order and pad it to length r with a new symbol; this is an injection into words of length r over $n+1$ symbols. Summing over t-tuples proves the claim, and iterating gives $W_m \le [T(1 + rm)^r]^m$.

Step 6: expected cost of $\mathcal{C}_2$. Fix an m-tuple of distinct regular edges. Its m^2 ordered pairs (e_i, e_j) are pairwise distinct, hence sampled independently, so the tuple contributes to $\mathcal{C}_2$ with probability $(1 - \pi)^{m^2} \le e^{-\pi m^2} \le 2^{-\pi m^2}$. (This is the only use of independence.) By linearity of expectation, again counting with multiplicity,

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le W_m\, 2^{-\pi m^2} \le [T(1 + rm)^r]^m\, 2^{-\pi m^2}.$$

Now $1+rm \le (h+1)2^{12h} \le 2^{13h}$ by $r \le h$ and $h+1 \le 2^h$, so $T(1+rm)^r \le 2^{4h+13rh}$. Also $\pi m = 2^{4h}$ and

$$4h + 13rh + 1 \le 5h^2 + 13h^2 = 18h^2 \le 18 \cdot 2^{2h} \le 2^{4h},$$

using $4h+1 \le 5h^2$, $r \le h$, $h \le 2^h$, and $18 \le 2^{2h}$ (as $h \ge 64$). Therefore

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le 2^{m(4h+13rh-\pi m)} \le 2^{-m}. \tag{2.4}$$

Step 7: conclusion. By (2.1), (2.3) and (2.4), $\mathbb{E}[c_\rho(\mathcal{G})] \le 2^{2-2h} + 2^{-m} \le 2 \cdot 2^{-h} \le 3 \cdot 2^{-h}$, since $h \ge 2$ and $m \ge h$. The sample space is finite, so some outcome has cost at most the expectation, and by Step 4 its $\mathcal{G}$ is a cover. □

3 Weights for exceptional sets

The next proposition gives a cover at a fixed fraction of the density. It is used in Section 4 with $q = Lp$.

Proposition 3.1. *Let $L = 2^{70}$. For every integer $N \ge 1$, every $q \in (0,1)$, and every $\mathcal{F} \subseteq 2^{[N]}$ satisfying $\mu_q(\mathcal{F}) \ge 1/2$, the family $E_{32}(\mathcal{F})$ is (q/L)-small.*

Proof. Plan. Define weights $w(U)$ (Step 1); prove a global bound (Step 2) and a local bound on every exceptional set (Step 3); then build the cover from one family of heavy sets and one application of Lemma 2.1 per bin (Steps 4–7).

Step 1: weights. Identify subsets of $[N]$ with indicator vectors and write $f = \mathbf{1}_{\mathcal{F}}$. For $U \subseteq [N]$ define

$$b(U) = (1-q)^{|U|} \sum_{z \in \{0,1\}^U} (-1)^{|z|}\, \mathbb{E}[f(z,Y)], \qquad w(U) = b(U)^2, \tag{3.1}$$

where $|z| = \sum_{i \in U} z_i$, the vector Y has independent Bernoulli-q coordinates on $[N] \setminus U$, and (z,Y) equals z on U and Y off U. In particular $b(\varnothing) = \mu_q(\mathcal{F}) \ge 1/2$.

Step 2: global bound. We claim

$$\sum_{U \subseteq [N]} q^{|U|} w(U) \le 1. \tag{3.2}$$

Let $X \sim \mu_q$. The functions

$$\chi_U(x) = \prod_{i \in U} \frac{q - x_i}{\sqrt{q(1-q)}}, \qquad U \subseteq [N],$$

form an orthonormal basis of the real functions on $\{0,1\}^N$ under μ_q (the biased Fourier basis [7, Section 8.4]): each factor has mean 0 and variance 1, the coordinates are independent, and there are 2^N functions. With $\widehat{f}(U) = \mathbb{E}[f(X)\chi_U(X)]$, Parseval's identity gives $\sum_U \widehat{f}(U)^2 = \mathbb{E}[f(X)^2] = \mu_q(\mathcal{F}) \le 1$. To relate $\widehat{f}$ to b, condition on the restriction of X to U:

$$\widehat{f}(U) = \sum_{z \in \{0,1\}^U} \Big(\prod_{i \in U} \frac{\mathbb{P}(X_i = z_i)(q - z_i)}{\sqrt{q(1-q)}} \Big) \mathbb{E}[f(z,Y)].$$

Both cases $z_i = 0, 1$ give $\mathbb{P}(X_i = z_i)(q - z_i) = q(1-q)(-1)^{z_i}$ (namely $(1-q)q$ and $q(q-1)$). Hence

$$\widehat{f}(U) = [q(1-q)]^{|U|/2} \sum_{z \in \{0,1\}^U} (-1)^{|z|}\, \mathbb{E}[f(z,Y)] = \left(\frac{q}{1-q}\right)^{|U|/2} b(U).$$

Squaring, $q^{|U|}w(U) = (1-q)^{|U|}\widehat{f}(U)^2 \le \widehat{f}(U)^2$; summing over U gives (3.2).

Step 3: local bound. Fix $S \in E_{32}(\mathcal{F})$. We claim

$$\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \ge 2^{-32}. \tag{3.3}$$

Consider arrays $x \in \{0,1\}^{32\times N}$ with rows $x_1, \dots, x_{32} \in \{0,1\}^N$ (subsets of $[N]$) and columns $x^{(1)}, \dots, x^{(N)} \in \{0,1\}^{32}$. For $y \in \{0,1\}^{32}$ with $|y| = \sum_j y_j$, set

$$P(y) = q^{|y|}(1-q)^{32-|y|}, \qquad Q(y) = P(y) - (1-q)^{32}(-1)^{|y|}, \qquad \nu_S(x) = \prod_{i\notin S} P(x^{(i)}) \prod_{i\in S} Q(x^{(i)}).$$

(Q is the source's R; R denotes random sets in Section 4.)

(a) Vanishing. We have $\sum_x \nu_S(x) \prod_{j=1}^{32} f(x_j) = 0$, term by term. Indeed $Q(0,\dots,0) = 0$, so if $\nu_S(x) \neq 0$ then every column in S contains a 1, that is, $S \subseteq x_1 \cup \cdots \cup x_{32}$. Since $S \in E_{32}(\mathcal{F})$, some row is not in $\mathcal{F}$, and $\prod_j f(x_j) = 0$. (The signs of ν_S play no role.)

(b) Expansion. Expand $\prod_{i\in S} Q(x^{(i)})$ by choosing, in each column $i \in S$, either $P(x^{(i)})$ or $-(1-q)^{32}(-1)^{|x^{(i)}|}$; let $U \subseteq S$ be the set of columns where the second term is chosen. Then

$$\nu_S(x) = \sum_{U\subseteq S} (-1)^{|U|}(1-q)^{32|U|} \prod_{i\in U} (-1)^{|x^{(i)}|} \prod_{i\notin U} P(x^{(i)}).$$

Each factor splits over the 32 rows: $(1-q)^{32|U|} = \prod_j (1-q)^{|U|}$, $\prod_{i\in U}(-1)^{|x^{(i)}|} = \prod_j (-1)^{\sum_{i\in U} x_{ji}}$, and $\prod_{i\notin U} P(x^{(i)}) = \prod_j \prod_{i\notin U} q^{x_{ji}}(1-q)^{1-x_{ji}}$. So $\nu_S(x) = \sum_{U\subseteq S}(-1)^{|U|}\prod_{j=1}^{32} g_U(x_j)$ with

$$g_U(u) = (1-q)^{|U|}(-1)^{\sum_{i\in U} u_i} \prod_{i\notin U} q^{u_i}(1-q)^{1-u_i} \qquad (u \in \{0,1\}^N).$$

The rows of x range independently over $\{0,1\}^N$, so $\sum_x \prod_j g_U(x_j) f(x_j) = \left(\sum_u g_U(u) f(u)\right)^{32}$. Writing $u = (z, y)$ with z on U and y off U, the sum over y against the Bernoulli masses is $\mathbb{E}[f(z, Y)]$, so $\sum_u g_U(u) f(u) = b(U)$ by (3.1). With (a),

$$0 = \sum_{U\subseteq S} (-1)^{|U|} b(U)^{32}. \tag{3.4}$$

(c) Conclusion. Isolating $U = \varnothing$ and using the triangle inequality,

$$2^{-32} \le b(\varnothing)^{32} \le \sum_{\varnothing\neq U\subseteq S} |b(U)|^{32} = \sum_{\varnothing\neq U\subseteq S} w(U)^{16}.$$

Step 4: heavy sets. Set $\rho = q/L$ and

$$\mathcal{G}_0 = \{U \subseteq [N] : U \neq \varnothing,\ w(U) \ge 2^{-64|U|}\}.$$

For $U \in \mathcal{G}_0$ with $|U| = r \ge 1$, $\rho^r = 2^{-6r} q^r 2^{-64r} \le 2^{-6r} q^r w(U) \le 2^{-6} q^r w(U)$, so by (3.2)

$$c_\rho(\mathcal{G}_0) \le 2^{-6} \sum_U q^{|U|} w(U) \le 2^{-6}. \tag{3.5}$$

Step 5: bins. For integers $r \ge 1$ and h (of any sign), let

$$\mathcal{H}_{r,h} = \{U \subseteq [N] : |U| = r,\ U \notin \mathcal{G}_0,\ 2^{-(h+1)} < w(U) \le 2^{-h}\}.$$

Every nonempty $U \notin \mathcal{G}_0$ with $w(U) > 0$ lies in exactly one bin. A nonempty bin has $h \geq 64r$: a member satisfies $2^{-(h+1)} < w(U) < 2^{-64r}$, so $h + 1 > 64r$. By (3.2),

$$|\mathcal{H}_{r,h}|\, q^r 2^{-(h+1)} < \sum_{U \in \mathcal{H}_{r,h}} q^r w(U) \leq 1,$$

and $\rho^r \leq q^r$ because $L \geq 1$; hence $|\mathcal{H}_{r,h}|\rho^r \leq 2^{h+1}$. So Lemma 2.1 applies with $X = [N]$ and $\mathcal{H} = \mathcal{H}_{r,h}$; its hypotheses $1 \leq r \leq h$ and $h \geq 64$ follow from $h \geq 64r \geq 64$. It gives a family $\mathcal{G}_{r,h}$ with $c_\rho(\mathcal{G}_{r,h}) \leq 3 \cdot 2^{-h}$ that covers every set containing at least 2^{12h} members of $\mathcal{H}_{r,h}$.

Step 6: the cover and its cost. Let $\mathcal{G}$ be the union of $\mathcal{G}_0$ and the $\mathcal{G}_{r,h}$ over the nonempty bins (finitely many, since $[N]$ is finite). As each nonempty bin has $h \geq 64r$,

$$c_\rho(\mathcal{G}) \leq 2^{-6} + 3 \sum_{r \geq 1} \sum_{h \geq 64r} 2^{-h} = \frac{1}{64} + 3 \sum_{r \geq 1} 2^{1-64r} = \frac{1}{64} + \frac{6}{2^{64} - 1} < \frac{1}{2}. \tag{3.6}$$

Step 7: $\mathcal{G}$ covers $E_{32}(\mathcal{F})$. Suppose $S \subseteq [N]$ contains no member of $\mathcal{G}$. Then S contains no member of $\mathcal{G}_0$ and, for each nonempty bin, fewer than 2^{12h} members of $\mathcal{H}_{r,h}$ (otherwise it would contain a member of $\mathcal{G}_{r,h}$). So every nonempty $U \subseteq S$ has $w(U) = 0$ or lies in a bin, where $w(U)^{16} \leq 2^{-16h}$. Therefore

$$\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \leq \sum_{r \geq 1} \sum_{h \geq 64r} 2^{12h} \cdot 2^{-16h} = \sum_{r \geq 1} \frac{16}{15}\, 2^{-256r} = \frac{16}{15(2^{256} - 1)} < 2^{-32}. \tag{3.7}$$

By (3.3), $S \notin E_{32}(\mathcal{F})$. Contrapositively, every member of $E_{32}(\mathcal{F})$ contains a member of $\mathcal{G}$; with (3.6), $E_{32}(\mathcal{F})$ is ρ-small. □

4 Returning to the original density

Proof of Theorem 1.1. Plan. Build $L = 2^{70}$ dependent random sets, each with law exactly μ_p, whose union has density Lp (or is all of $[N]$). A union bound puts all of them in $\mathcal{D}$ simultaneously with high probability. Then apply Proposition 3.1 to unions of L members of $\mathcal{D}$. (The source's Case 2 parameter t is called θ here.)

Let $L = 2^{70}$, so $k = 32L$, and assume $\mu_p(\mathcal{D}) \geq 1 - 1/k$. If $R_1, \ldots, R_L$ are random subsets of $[N]$ on one probability space, each with law μ_p, then, with no independence assumption,

$$\mathbb{P}(R_j \in \mathcal{D} \text{ for all } 1 \leq j \leq L) \geq 1 - \sum_{j=1}^{L} \mathbb{P}(R_j \notin \mathcal{D}) \geq 1 - \frac{L}{k} = \frac{31}{32}. \tag{4.1}$$

Case 1: $Lp < 1$. Independently for each $i \in [N]$, choose a label $J_i \in \{0, 1, \ldots, L\}$ with

$$\mathbb{P}(J_i = 0) = 1 - Lp, \qquad \mathbb{P}(J_i = j) = p \quad (1 \leq j \leq L),$$

and put $R_j = \{i : J_i = j\}$, so each element goes to at most one row. For fixed j, the events $\{i \in R_j\}$ are independent over i, each of probability p; so R_j has law μ_p. The union $R = R_1 \cup \cdots \cup R_L = \{i : J_i \neq 0\}$ has independent coordinates, each present with probability $q := Lp \in (0, 1)$; so R has law μ_q.

Define $\mathcal{F} = \{D_1 \cup \cdots \cup D_L : D_1, \ldots, D_L \in \mathcal{D}\}$. On the event in (4.1), $R \in \mathcal{F}$, hence

$$\mu_q(\mathcal{F}) = \mathbb{P}(R \in \mathcal{F}) \geq \frac{31}{32} \geq \frac{1}{2}.$$

The unions of 32 members of $\mathcal{F}$ are exactly the unions of $32L$ members of $\mathcal{D}$: expand each member of $\mathcal{F}$, or conversely group a $32L$-tuple into 32 blocks of length L. Thus

$$E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D}) = E_k(\mathcal{D}).$$

Proposition 3.1 shows that this family is (q/L)-small, and $q/L = p$.

Case 2: $Lp \geq 1$. Set

$$\theta = \frac{p - 1/L}{1 - 1/L},$$

so $\theta \in [0, 1)$ because $1/L \leq p < 1$. Independently for each $i \in [N]$: choose one row uniformly among the L rows and put i in it; then put i into each other row independently with probability θ. For fixed j,

$$\mathbb{P}(i \in R_j) = \frac{1}{L} + \left(1 - \frac{1}{L}\right)\theta = p,$$

and these events are independent over i, so R_j has law μ_p; moreover $R_1 \cup \cdots \cup R_L = [N]$ in every outcome. By (4.1), some outcome has all $R_j \in \mathcal{D}$. Repeating these L sets 32 times gives k members of $\mathcal{D}$ whose union $[N]$ contains every S. Hence $E_k(\mathcal{D}) = \varnothing$, which the empty family covers at cost 0. (At the boundary $Lp = 1$, $\theta = 0$.) □

5 Verification checklist

1. *Extension bound* (2.2) (Lemma 2.1, Step 2). Regularity is used exactly to get $d(e \cap B) \leq T$. This bound drives both (2.3) and the recursion in Step 5.
2. *Residual cost* (Lemma 2.1, Step 6). The m^2 pairs of a tuple are distinct, giving probability $(1 - \pi)^{m^2}$; then the arithmetic $4h + 13rh + 1 \leq 2^{4h} = \pi m$.
3. *Coefficient identity* (Proposition 3.1, Step 2). $\widehat{f}(U) = (q/(1-q))^{|U|/2} b(U)$, and the discarded factor $(1-q)^{|U|} \leq 1$ goes in the right direction.
4. *Signed identity* (3.4) (Step 3). $Q(0, \ldots, 0) = 0$ gives termwise vanishing; the row factorisation produces $b(U)^{32}$; $b(\varnothing) \geq 1/2$.
5. *Bins* (Steps 5–7). Nonempty bins have $h \geq 64r$; the sums (3.6) and (3.7); the threshold 2^{12h} is the same in Lemma 2.1 and in Step 7.
6. *Coupling* (Section 4). Each row has law exactly μ_p; the union has law μ_{Lp} (Case 1) or equals $[N]$ (Case 2); $E_{32}(\mathcal{F}) = E_k(\mathcal{D})$ and $q/L = p$.
7. *Corollary 1.2 only.* It depends on cited statements that are not checked here (see the boxed note); Theorem 1.1 does not use it.

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